

2520080061

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WEEK-4

BOOKFLOW INTELLIGENT

CATALOG & API BLUEPRINT

The screenshot displays the MySQL Workbench interface. The left sidebar shows the 'SCHEMAS' panel with a tree view containing '2520080061_pranay', 'joins_db', 'klu', 'pranay_2520080061', and 'sys'. The main editor window shows a SQL script with the following queries:

```
1 • DROP DATABASE IF EXISTS bookflow_db;
2   Execute the selected portion of the script or everything, if there is no selection
3 • CREATE DATABASE bookflow_db;
4
5 • USE bookflow_db;
6
7 • SELECT DATABASE();
8
9 • CREATE TABLE Books
10  (
11    book_id INT AUTO_INCREMENT PRIMARY KEY,
12    title VARCHAR(200) NOT NULL,
13    author VARCHAR(150) NOT NULL,
14    genre VARCHAR(100),
15    description TEXT,
16    published_year INT
17  );
18 • DESCRIBE Books;
19
20 • INSERT INTO Books
21  (
22    title,
23    author,
```

The bottom panel shows the 'Output' tab with 'Action Output' selected. It displays a table with columns '#', 'Time', 'Action', and 'Message'.

#	Time	Action	Message
✓ 1	12:23:07	DROP DATABASE IF EXISTS bookflow_db	1 row(s) affected
✓ 2	12:23:16	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	12:23:28	USE bookflow_db	0 row(s) affected

```

7 • SELECT DATABASE();
8
9 • CREATE TABLE Books
10 (
11     book_id INT AUTO_INCREMENT PRIMARY KEY,
12     title VARCHAR(200) NOT NULL,
13     author VARCHAR(150) NOT NULL,
14     genre VARCHAR(100),
15     description TEXT,

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	DATABASE()
▶	bookflow_db

Result 1 x

Output

Action Output

#	Time	Action	Message
✓ 1	12:23:07	DROP DATABASE IF EXISTS bookflow_db	1 row(s) affected
✓ 2	12:23:16	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	12:23:28	USE bookflow_db	0 row(s) affected
✓ 4	12:24:10	SELECT DATABASE() LIMIT 0, 2000	1 row(s) returned

PART 2 — Create the Books table

```

10 (
11     book_id INT AUTO_INCREMENT PRIMARY KEY,
12     title VARCHAR(200) NOT NULL,
13     author VARCHAR(150) NOT NULL,
14     genre VARCHAR(100),
15     description TEXT,
16     published_year INT
17 );
18 • DESCRIBE Books;
19
20 • INSERT INTO Books
21 (

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
▶ book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 5	12:24:42	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	0 row(s) affected
✗ 6	12:24:43	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	Error Code: 1050. Table 'books' already exists
✗ 7	12:24:53	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	Error Code: 1050. Table 'books' already exists
✓ 8	12:24:57	DESCRIBE Books	6 row(s) returned

PART 3 — Insert the books

```
86 'Harry Potter and the Sorcerers Stone',
87 'J.K. Rowling',
88 'Fantasy',
89 'A young wizard begins his magical education.',
90 1997
91 ),
92 (
93 'Twenty Thousand Leagues Under the Seas',
94 'Jules Verne',
95 'Adventure',
96 'A fantastic underwater adventure aboard a mysterious submarine.',
97 1870
98 );
99 • SELECT * FROM Books;
100 • ALTER TABLE Books
```

Output

#	Time	Action	Message
6	12:24:43	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	Error Code: 1050. Table 'books' already exists
7	12:24:53	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	Error Code: 1050. Table 'books' already exists
8	12:24:57	DESCRIBE Books	6 row(s) returned
9	12:25:55	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```
93 'Twenty Thousand Leagues Under the Seas',
94 'Jules Verne',
95 'Adventure',
96 'A fantastic underwater adventure aboard a mysterious submarine.',
97 1870
98 );
99 • SELECT * FROM Books;
100 • ALTER TABLE Books
101 ADD COLUMN embedding JSON;
102 • DESCRIBE Books;
103
104 • UPDATE Books
105 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
106 WHERE book_id = 1;
107
```

Result Grid

book_id	title	author	genre	description
1	The Martian	Andy Weir	Science Fiction	An astronaut struggles to survive alone on Mars.
2	Dune	Frank Herbert	Science Fiction	A story involving planets, politics, survival and s..
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	A humorous adventure across space and alien ...
4	Pride and Prejudice	Jane Austen	Romance	A classic story about relationships and social ex...

Books 3 x

Output

#	Time	Action	Message
7	12:24:53	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT N...	Error Code: 1050. Table 'books' already exists
8	12:24:57	DESCRIBE Books	6 row(s) returned
9	12:25:55	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplic...
10	12:26:40	SELECT * FROM Books LIMIT 0, 2000	10 row(s) returned

PART 4 — Add the embedding column

```
99 • SELECT * FROM Books;
100 • ALTER TABLE Books
101   ADD COLUMN embedding JSON;
102 • DESCRIBE Books;
103
```

Context Help Snippet

Output

Action Output

#	Time	Action	Message
✓ 8	12:24:57	DESCRIBE Books	6 row(s) returned
✓ 9	12:25:55	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 10	12:26:40	SELECT * FROM Books LIMIT 0, 2000	10 row(s) returned
✓ 11	12:27:07	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

```
102 • DESCRIBE Books;
103
104 • UPDATE Books
```

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	

Result 4 x

Read Only Context Help Snippet

Output

Action Output

#	Time	Action	Message
✓ 9	12:25:55	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 10	12:26:40	SELECT * FROM Books LIMIT 0, 2000	10 row(s) returned
✓ 11	12:27:07	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
✓ 12	12:27:30	DESCRIBE Books	7 row(s) returned

PART 5 — Add the mock vectors

Limit to 2000 rows

Automatic completion manually generated

```
104 • UPDATE Books
105     SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)
106     WHERE book_id = 1;
107
108 • UPDATE Books
109     SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)
110     WHERE book_id = 2;
111
112 • UPDATE Books
113     SET embedding = JSON_ARRAY(0.80, 0.70, 0.30)
114     WHERE book_id = 3;
115
116 • UPDATE Books
117     SET embedding = JSON_ARRAY(0.10, 0.20, 0.90)
118     WHERE book_id = 4;
119
120 • UPDATE Books
121     SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
122     WHERE book_id = 5;
123
124 • UPDATE Books
125     SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
126     WHERE book_id = 6;
```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 15	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 16	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

```
144 • SELECT
145     book_id,
146     title,
147     embedding
148 FROM Books
149 ORDER BY book_id;
150
151 • SELECT
```

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]

Books 5 x

Apply

Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 16	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 17	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 18	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	12:28:34	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 2000	10 row(s) returned

PART 6 — L2 Distance Search

```
169      POW(  
170          CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20,  
171          2  
172      ),  
173      4  
174  ) AS l2_distance  
175  
176  
177  FROM Books  
178  
179  WHERE embedding IS NOT NULL  
180  
181  ORDER BY l2_distance ASC  
182  
183  LIMIT 3;
```

Result Grid

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
	2	Dune	Frank Herbert	Science Fiction	0.0866
	6	Foundation	Isaac Asimov	Science Fiction	0.1581

Result 6 x

Output

Action Output

#	Time	Action
✓ 17	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5
✓ 18	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 19	12:28:34	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 2000
✓ 20	12:29:16	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSO...

PART 7 — Cosine Similarity

```

230      POW(0.20, 2)
231    )
232  ),
233
234    4
235
236  ) AS cosine_similarity
237
238  FROM Books
239
240  WHERE embedding IS NOT NULL
241
242  ORDER BY cosine_similarity DESC
243
244  LIMIT 3;

```

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	0.9939

#	Time	Action
✓ 18	12:28:02	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 19	12:28:34	SELECT book_id, title, embedding FROM Books ORDER BY book_id LIMIT 0, 2000
✓ 20	12:29:16	SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSO...
✓ 21	12:29:40	SELECT book_id, title, author, genre, ROUND((CAST(JSON_EXTRACT(embeddin...